

A Connection between Optimal Inference and Thermodynamic Efficiency in Brains

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Abstract

This paper establishes a formal connection between thermodynamic efficiency and optimal statistical inference. Biological neural networks can be understood to meet two competing demands: accurately making inferences in an evolving environment and minimizing energy consumption. By mapping the information-theoretic thermodynamic bounds of Still et al. (2012) onto continuous Linear Gaussian dynamics, we show that the discrete-time Kalman filter is consistent with minimizing thermodynamic dissipation under these assumptions. Numerical simulations of a visual tracking task demonstrate that suboptimal inference increases predictive entropy, which corresponds to increased thermodynamic dissipation.

1 Introduction

A brain situated in a rich environment will be exposed to many external driving signals, and therefore the dynamics of the brain will be a function of these signals. Therefore, the brain will retain some information or memory about the environment's signals. If these external signals are correlated in some way, then the brain can also be interpreted to implicitly contain a prediction of future signals simply by virtue of being exposed to current signals.

Still et al. [1] provide a non-equilibrium thermodynamic framework in which we can differentiate between the predictive and the useless information an object retains about the signals of its environment. This difference is then demonstrated to be a lower bound on the energy wasted for predicting the outside signal. On the other hand side, the Kalman filter [2] is a well known optimal statistical estimation of a latent dynamic variable. Due to the information theoretic terms in Still et al.'s framework, we can relate the Kalman filter to thermodynamics. In this paper we will demonstrate how in a linear Gaussian environment, a physical system operating at the absolute lower bound of thermodynamic dissipation is mathematically equivalent to a Kalman filter. If we consider neuronal populations of the brain as encoding computations, then this optimality of the Kalman filter seems particularly relevant in two ways: brains should energy-efficiency [3], and brains should provide optimal inference of the environment [4]. Prima facie these two objectives seem at odds. However, in a Linear Gaussian

environment we can demonstrate that both of these imperatives are fulfilled by the Kalman filter. We will only consider inference, and not control in this paper. As such, we will first specify the thermodynamics of prediction framework in section 2, then translate that over into a linear Gaussian environment in section 3. In section 4, the most crucial part of the paper is demonstrated: how maximizing thermodynamic efficiency is equivalent to maximizing optimal statistical inference in section . The relevance of this dual optimality is discussed for brains then in section 5. Further, some simulations are provided to illustrate the theoretical results in section 6. Finally, important caveats and possible extensions are discussed in 7. Lengthier and detailed derivations are relegated to the appendix.

2 Thermodynamics of Prediction model

In Still's model [1], we have:

- Physical states of the system s
- Discrete time t
- Signals to the system x , which disequilibrate the system over a trajectory or protocol $x_0, \dots, x_\tau$
- Markovian state transition distribution $p(s_t | s_{t-1}, x_t)$
- External drive joint distribution $P_X = p(x_0, \dots, x_\tau)$, which can be non-markovian.
- Inverse temperature $\beta = \frac{1}{kT}$, with heat bath at temperature T

The system is in thermodynamic equilibrium at $t = 0$. The energy of the system at any time is given by $E(s, x_t)$. The equilibrium free energy is given by: $F_t = F[x_t]$. The equilibrium distribution is then:

$$p_{eq}(s|x_t) := e^{-\beta(E(s, x_t) - F_t)} \quad (1)$$

The probability of a specific trajectory of the system is:

$$P_{S|X} = p_{eq}(s_0|x_0) \prod_{t=1}^{\tau} p(s_t | s_{t-1}, x_t) \quad (2)$$

And the joint probability of a trajectory and signals is:

$$P_{X,S} = p(x_0)p_{eq}(s_0|x_0) \prod_{t=1}^{\tau} p(x_t | x_0, \dots, x_{t-1})p(s_t | s_{t-1}, x_t) \quad (3)$$

The non-equilibrium free energy is then the expected energy minus temperature times entropy:

$$F_{neq}[p(s|x)] := \langle E(s, x) \rangle_{p(s|x)} + kT \langle \ln p(s|x) \rangle_{p(s|x)} \quad (4)$$

Over a trajectory them, the nonequilibrium free energy change is:

$\Delta F_{neq} := F_{neq}[p(s_\tau|x_\tau)] - F_{neq}[p(s_0|x_0)]$. As a system evolves over time, it incorporates

information about the signal like a memory:

$$I_{mem}(t) := I(s_t, x_t) = \left\langle \frac{p(s_t, x_t)}{p(s_t)p(x_t)} \right\rangle_{p(s_t, x_t)} \quad (5)$$

The useful predictive information in the system about the signal is:

$$I_{pred}(t) := I(s_t, x_{t+1}) = H(x_{t+1}) - H(x_{t+1}|s_t) \quad (6)$$

The difference between these two is called ‘nostalgia’ or information with no predictive use: $I_{mem} - I_{pred}$.

As the external signal changes, work is done:

$$W[s_{t-1}, x_{t-1} \rightarrow x_t] := E(s_{t-1}, x_t) - E(s_{t-1}, x_{t-1}) \quad (7)$$

The average work irretrievably lost i.e dissipated over a trajectory of the system :

$$\langle W_{diss} \rangle_{P_{S|X}} := \langle W \rangle_{P_{S|X}} - \Delta F_{neq} \quad (8)$$

The nostalgia information is then exactly proportional to the average work dissipated by the system as the signal transitions. Over a whole trajectory or protocol, the ‘nostalgia’ is a lower bound on the dissipated work: (See appendix A for details on this derivation):

$$\begin{aligned} I_{mem}(t) - I_{pred}(t) &= \beta \langle W_{diss}(x_t \rightarrow x_{t+1}) \rangle \\ I_{mem} - I_{pred} &\leq \beta \langle W_{diss} \rangle \end{aligned} \quad (9)$$

3 Linear Gaussian Environment

The Linear Gaussian (LG) setting is the simplest environment for estimating the dynamics of a noisy latent variable [2], [5]. For example, consider a hunting cheetah tracking the position of a gazelle running through shrubbery. The following list provides all variables we use in this setting for such an inference problem:

- The true objective state of some latent variable $x_t \in \mathbb{R}^n$.
- States of the system performing inference $s_t \in \mathbb{R}^m$. For the purposes of this paper however, this is amended later.
- Measurements of the latent variable $y_t \in \mathbb{R}^m$.
- The system’s prediction of the latent state before sensory input: $\hat{x}_{t|t-1} \in \mathbb{R}^n$, and the updated estimate after incorporating measurement: $\hat{x}_{t|t} \in \mathbb{R}^n$
- The transition-of-states matrix $A \in \mathbb{R}^{n \times n}$, and the observation matrix $C \in \mathbb{R}^{m \times n}$ mapping state space to measurement space
- Process noise in the evolution of x called $\omega \sim N(0, Q)$, where the process noise covariance $Q \in \mathbb{R}^{n \times n}$ is *strictly* positive definite
- Measurement noise due to thermal noise $v \sim N(0, R)$, where the measurement noise covariance $R \in \mathbb{R}^{m \times m}$ is *strictly* positive definite

- Predictive error covariance $\Sigma_{t|t-1} \in \mathbb{R}^{n \times n}$ which is positive semi-definite, and posterior error covariance $\Sigma_{t|t} \in \mathbb{R}^{n \times n}$ which is positive semi-definite
- The Kalman gain $K \in \mathbb{R}^{n \times m}$

To bridge Still's framework to the LG setting, we explicitly define the environmental state x and system state s as continuous random variables. While information-theoretic bounds are often formulated using discrete spaces for notational convenience, we can seamlessly extend these to continuous spaces via differential Shannon entropies.

The external drive is given by:

$$x_{t+1} = Ax_t + \omega_t \quad (10)$$

Where A is the state transition matrix, and $\omega \sim N(0, Q)$ is Gaussian process noise with covariance matrix Q .

The measurements y of x are noisy due to thermal fluctuations:

$$y_t = Cx_t + v_t \quad (11)$$

Where C is the observation matrix, and $v \sim N(0, R)$ is thermal noise from the heat bath, and R is the measurement noise covariance.

Given the LG setting, the optimal prediction $\hat{x}$ of x is derived by the recursive update:

$$\hat{x}_t = A\hat{x}_{t-1} + K(y_t - CA\hat{x}_{t-1}) \quad (12)$$

Where K is the Kalman gain, which we solve for later. This factor determines how much the current predicted state is adjusted based on the sensory prediction error. We can interpret this as the synaptic weights for a brain.

As in Still et al., the system must model its environment. So we will replace the system state with an estimate of the environment. The system's state (membrane potentials or firing rates) must serve as the optimal sufficient statistic of the environment, and provides an estimate of the environment's state. In our Gaussian setting, the system's state therefore is equivalent to the tuple of the average estimate of the environment and the internal uncertainty about the environment.

$$s_t \equiv (\hat{x}_t, \Sigma_t) \quad (13)$$

Where Σ_t is the error covariance matrix, which updates recursively alongside the mean to track the width of the posterior estimate:

$$\Sigma_t = (I - KC)(A\Sigma_{t-1}A^T + Q) \quad (14)$$

The system's dynamics are then accordingly given by $p(s_t|s_{t-1}, y_t)$, and are to be optimized.

4 Relation to Kalman Filter

Still et al. have established that a thermodynamically optimal system minimizes its dissipated work by minimizing the nostalgia $I_{mem} - I_{pred}$. Assuming that a system is physically constrained to retain a certain amount of memory about its environment, minimizing dissipation boils down to maximizing the predictive mutual information:

$$I_{pred}(t) = H(x_{t+1}) - H(x_{t+1}|s_t) \quad (15)$$

The system has no effect on the first term, the signal's entropy, as we are solely concerned with inference and not control. Therefore the system can only minimize the latter term, the conditional entropy. The corresponding distribution of $p(x_{t+1}|s_t)$ is Gaussian, as given by our LG setting. So we can write the conditional entropy term as [6]:

$$H(x_{t+1}|s_t) = \frac{k}{2} \ln(2\pi e) + \frac{1}{2} \ln |\Sigma|_{t+1|t} \quad (16)$$

Where $|\Sigma|_{t+1|t}$ is the determinant of the predictive error covariance. To minimize the conditional entropy therefore is to minimize the determinant or 'volume' of the error covariance matrix.

How does the predictive covariance update? In our setting, it evolves according to the objective dynamics of the external drive and the process noise:

$$\Sigma_{t+1|t} = A\Sigma_{t|t}A^T + Q \quad (17)$$

So the only term the system can minimize is $\Sigma_{t|t}$, the current estimate uncertainty.

Now we finally relate the predicted environmental state $\hat{x}_t$ to the predictive covariance, making heavy use of the Gaussian linear setting and the relationship between measurement y and latent state x . The posterior error covariance is given by the expected square error of the estimate and can be rearranged as follows (see appendix B for details):

$$\begin{aligned} \Sigma_{t|t} &= \mathbb{E}_{p(x_t, v_t|s_{t-1})} [(x_t - \hat{x}_{t|t})(x_t - \hat{x}_{t|t})^T] \\ \Sigma_{t|t}(K) &= (I - KC)\Sigma_{t|t-1}(I - KC)^T + K RK^T \end{aligned} \quad (18)$$

The first term represents the reduction in prior uncertainty achieved by making a measurement. The second term is a penalty of absorbing the thermal noise in measurement.

Our objective now is to find the following optimizing gain:

$$K^* = \arg \min_K \log |\Sigma_{t|t}(K)| \quad (19)$$

Analytically minimizing the determinant, however, is a tedious exercise. We will make an *Ansatz* that the K which minimizes the trace will also minimize

the determinant since both terms of $\Sigma_{t|t}$ in (18) are positive semi-definite; We aim to show that the trace-minimizing K^* provides global optimality via a Loewner order. We therefore minimize the analytically tractable trace, call it $J = \text{tr}(\Sigma_{t|t})$. See Appendix C for exact optimization).

$$\begin{aligned} J &= \text{tr}(\Sigma_{t|t-1}) - 2\text{tr}(K C \Sigma_{t|t-1}) + \text{tr}(K(C \Sigma_{t|t-1} C^T + R) K^T) \\ \frac{dJ}{dK} &= -2\Sigma_{t|t-1} C^T + 2K(C \Sigma_{t|t-1} C^T + R) \stackrel{!}{=} 0 \\ K^* &= \Sigma_{t|t-1} C^T (C \Sigma_{t|t-1} C^T + R)^{-1} \end{aligned} \quad (20)$$

Where K^* is the optimal gain, used to update predictions $\hat{x}_{t|t}$. We can plug this K^* back into the equation for $\Sigma_{t|t}$:

$$\begin{aligned} \Sigma_{t|t}^* &= \Sigma_{t|t-1} - K^* C \Sigma_{t|t-1} - \Sigma_{t|t-1} C^T K^{*T} + K^* (C \Sigma_{t|t-1} C^T + R) K^{*T} \\ &= (I - K^* C) \Sigma_{t|t-1} \end{aligned} \quad (21)$$

This is the optimal gain K^* and error covariance $\Sigma_{t|t}^*$ as found in the Kalman filter update [5, pg. 129].

Now, what is the relationship of $\Sigma_{t|t}(K^*)$ to any alternative $\Sigma_{t|t}(K)$? Specifically, what is the definiteness of their difference?

We make use of the Loewner partial ordering, which is an ordering on positive semi-definite matrices [7, Ch. 7]. This ordering also gives the ordering of specific functionals of matrices, such as the determinant and the trace. So we need to demonstrate that $\Sigma_{t|t}(K^*)$ and $\Sigma_{t|t}(K)$ form such an ordering for any K .

We can make use of the fact that all terms of $\Sigma_{t|t}(K)$ are positive semi-definite, since $\Sigma_{t|t-1}$ is positive semi-definite. Crucially, R and Q are positive definite (since there is guaranteed thermal noise). We can use the solved-for formulation of K^* in (20) and rearrange to the following formulation:

$$\begin{aligned} S &:= C \Sigma_{t|t-1} C^T + R \\ \Sigma_{t|t}(K) &= \Sigma_{t|t}(K^*) + \underbrace{(K - K^*) S (K - K^*)^T}_{\text{PSD penalty}} \end{aligned} \quad (22)$$

So we can understand any $\Sigma_{t|t}(K)$ to be some positive semi-definite penalty away from the optimal $\Sigma_{t|t}(K^*)$. So since the difference $\Sigma_{t|t}(K) - \Sigma_{t|t}(K^*)$ is positive semi-definite, we can conclude that $\Sigma_{t|t}(K^*)$ is the bottom floor of the Loewner partial ordering. It is known that if in this ordering $\Sigma_{t|t}(K) \succeq \Sigma_{t|t}(K^*)$, then $\text{tr}(\Sigma_{t|t}(K)) \geq \text{tr}(\Sigma_{t|t}(K^*))$ and, most importantly, $|\Sigma_{t|t}(K)| \geq |\Sigma_{t|t}(K^*)|$. Note that since $\ln(\cdot)$ is a monotonic transformation, the minimizing argument doesn't shift. For details on this argument, see Appendix D.

So given the crucial assumptions of our LG setting, we can conclude the most important equivalence of this paper:

$$\arg \min_K \ln |\Sigma_{t|t}(K)| = \arg \min_K \text{tr}(\Sigma_{t|t}(K)) \quad (23)$$

5 Brains as thermodynamically efficient and statistically optimal

The demonstrated equivalence between optimal statistical inference and thermodynamic efficiency hints at a unified imperative for predictive biological systems such as a brain: on one hand side, it should provide good inference of the environment; On the other hand, it should remain resource efficient [8]. If we interpret the dynamic neuronal population codes as actually performing computations, then we can relate the thermodynamic cost of those computations directly to the quality of the brain’s predictive model of its environment.

In computational neuroscience, predictive processing is a dominant paradigm for sensory processing, which has been empirically substantiated. [4]. This sensory framework suggests that the brain does not simply absorb observational inputs. It actively creates a generative model of observations, and then updates this generative model based on errors relative to actual perceived observations. For continuous variables undergoing linear dynamics with Gaussian noise, the Kalman filter provides the optimal statistical estimation of a latent variable. To implement this neurologically, a population of neurons would need to encode the posterior estimate $\hat{x}_{t|t}$ and associated uncertainty $\Sigma_{t|t}$, and update these estimates via some biological analogue of the Kalman gain.

On the other hand side, what is the incentive to be energetically efficient? Although the human brain is roughly 2% of body weight, it accounts for 20% of the resting metabolism. The vast majority, ca. 80% , of the brain’s energy is expended on action potentials and post-synaptic activity [3]. Because the fundamental units of neural computation are so metabolically expensive, there is pressure to eliminate redundant or useless neural activity. As foundational work by Laughlin [9] demonstrated, neurons physically adapt their response functions to match the cumulative probability distribution of environmental inputs, thereby maximizing the information transmitted per unit of response range.

In Still’s framework, this useless information constitutes nostalgia, which bounds the amount of thermodynamic work dissipated as heat. We can therefore interpret a thermodynamic cost to a brain implementing a suboptimal gain K , i.e accruing past information which does not aid in predicting the external environmental state. A sub-optimal gain leads to larger posterior error covariance $\Sigma_{t|t}$ than necessary. This excess error can be interpreted to manifest itself as wasted metabolic energy.

It is important to note the difference in scale between Still et al.’s thermodynamic bound and biological energy costs. The nostalgia term, scaled by $k_B = 1.38 \times 10^{-23}$ Joules per Kelvin, operates many magnitudes below the metabolic cost of a single action potential, estimated by Attwell & Laughlin [3] at approximately 7×10^8 ATP molecules per spike, whereby one molecule of ATP provides roughly 8.3×10^{-20} Joules. Nevertheless, a proportional relationship may persist at a populational level: A suboptimal gain K maintains a broader posterior $\Sigma_{t|t}$, corresponding to higher predictive conditional entropy $H(x_{t+1}|s_t)$ and thus greater nostalgia. In a population code, broader uncertainty distributions could require sustained activity across more costly neurons to represent the same environmental state.

The thermodynamic optimality of the Kalman filter can be understood to underwrite

Barlow’s Efficient Coding Hypothesis [10]. This hypothesis posits that sensory neurons are adapted to represent the environment as population codes with maximal efficiency. Our results suggests that a neural population implementing a Kalman filter is simultaneously maximally thermodynamically efficient. With a fixed or insurmountable I_{mem} and trying to minimize I_{pred} our unified framework essentially posits the brain as solving a similar problem to the Information Bottleneck [11]. The past sensory data should be as compressed or minimized as possible, while maximizing the useful information for inference.

6 Simulations

To illustrate the theoretic results, we will implement a simulation on python.

Consider the task of visually tracking a point in a noisy environment; say a cheetah was chasing prey through shrubbery and keeps track of a gazelle’s position. The gazelle runs away roughly in a straight line. The gazelle’s position evolves according to linear dynamics, with some unpredictable jumps left and right given by Q . The cheetah further has thermal noise R in its sensory apparatus. The cheetah’s brain must estimate the true position of the gazelle.

We compare two estimators for the position of the target:

The optimal Kalman filter using K^* versus a suboptimal filter employing a fixed gain K' . We first set $K' < K^*$, which means the suboptimal gain trusts prior predictions too much and fails to adequately incorporate new evidence. We then examine $K' > K^*$, which means the suboptimal gain is too sensitive to new measurements and ‘forgets’ its prior prediction.

How exactly does the sub-optimality of K' manifest? We can compare the predicted trajectories for the two estimators for $\tau = 50$ time steps. On a statistical basis, we can consider the mean squared error cost function. On a thermodynamic basis, we consider the excessive non-predictive ‘nostalgia’ difference, whereby I_{mem} is fixed, so the estimators need to minimize I_{pred} :

$$MSE = \frac{1}{T} \sum_{t=1}^{\tau} (x_t - \hat{x}_{t|t})^2 \quad I_{pred} = H(x_{t+1}) - H(x_{t+1}|s_t) \quad (24)$$

We compare a ‘stubborn’ gain (Fig. 1) and an ‘eager’ gain (Fig. 2) versus the optimal gain. See appendix for code.

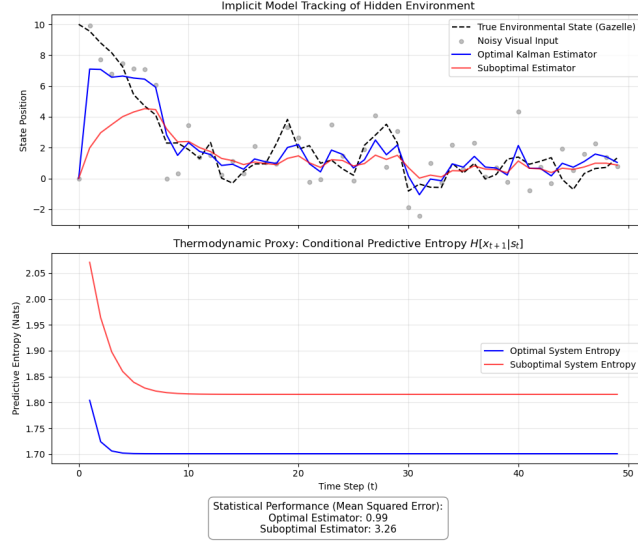


Fig. 1 Comparison performance of **Stubborn** $K' < K^*$ in tracking problem.

Parameters: $\tau = 50, A = 0.9, C = 1, Q = 1, R = 2$

$$K' = 0.2, K^* = \Sigma_{t|t-1} C^T (C \Sigma_{t|t-1} C^T + R)^{-1}$$

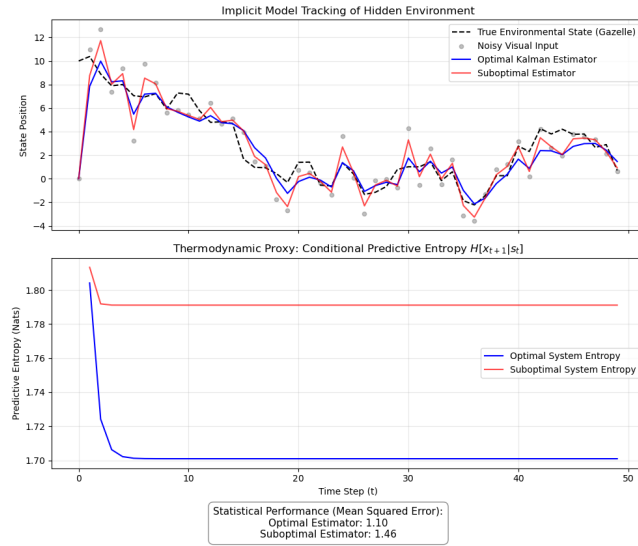


Fig. 2 Comparison performance of **Eager** $K' > K^*$ in tracking problem.

Parameters: $\tau = 50, A = 0.9, C = 1, Q = 1, R = 2$

$$K' = 0.8, K^* = \Sigma_{t|t-1} C^T (C \Sigma_{t|t-1} C^T + R)^{-1}$$

7 Conclusion

This paper has demonstrated a formal connection between optimal statistical inference and information-theoretic thermodynamic efficiency. By translating the generalised state variables of Still et al.’s framework into a continuous linear Gaussian setting, we showed that the discrete-time Kalman filter provides not merely statistically optimal inference, but also minimizes the thermodynamic dissipation bound in this setting. However, given the assumptions and physical scales involved, these results must be interpreted cautiously when applied to biological systems.

A primary critique of invoking the thermodynamics of computation is the mismatch in scale. The nostalgia term we minimized is simply a lower bound on the wasteful work. To make the information theoretic term commensurate to work, we multiply the nostalgia by $k_b T$, where $k_b = 1.3810^{-23} \frac{J}{K}$. However, while the lower bound is many magnitudes below the metabolic Joule cost of a single neural spike, there could be a proportional cost in an entire neuronal population.

Furthermore, our derivation makes use of the abstraction that neural population can be described Gaussian sufficient statistics $s_t \equiv (\hat{x}_t, \Sigma_t)$, abstracting away biological complexity.

While a brain operates many magnitudes above this k_B limit, a theoretical proportional bound remains. In a neural population code, excessive predictive entropy or ‘nostalgia’ requires maintaining broad distributions. This could lead to wasteful neural firing, potentially increasing metabolic cost at large scales.

Looking towards control, the Good Regulator Theorem (originally [12], more modern [13]) implies that any ‘good’ controller must embody a model of the system it regulates. Under Linear Quadratic Gaussian assumptions, optimal control permits the separation principle: State estimation (via Kalman filter) and control (via the Linear Quadratic Regulator) can be treated as independent problems. This separation principle is not in conflict with the Good Regulator Theorem – the controller still relies on a model – but it is a special artifact of the linear dynamics and Gaussian noise.

This separation breaks down in more realistic settings involving non-linear dynamics and signal-dependent noise. However, as Todorov [14] demonstrated, the breakdown of separation reveals a deeper equivalence between perception and action. In certain formulations, optimal control can be recast in inference-like terms. For example, when control costs are expressed as a KL-divergence from passive dynamics, the problem admits a probabilistic interpretation. This suggests a formal bridge to the Still et al. framework

A natural extension then could be to model a brain as an active Information Bottleneck [11]. An organism would compress its sensory history to not only predict the future, but to actively act on it. Developing this connection may provide a joint perspective on control, estimation, and efficiency.

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Appendix A Lower dissipated work bound

This is a summary on how equation (18) was derived in Still et al., the central lower bound relation for this paper.

Start with the definition of work, take expectation, and multiply by β :

$$\begin{aligned} W[s_t; x_t \rightarrow x_{t+1}] &:= E[s_t, x_{t+1}] - E[s_t, x_t] \\ \beta\langle W \rangle &= \beta\langle E[s_t, x_{t+1}] - E[s_t, x_t] \rangle_{p(s_t, x_t, x_{t+1})} \end{aligned} \quad (\text{A1})$$

Then by the definition of non-equilibrium free energy, take expectation in $p(x)$, multiply by β :

$$\begin{aligned} F_{neq} &:= \langle E[s, x] \rangle_{p(s|x)} + k_B T \langle \log p(s|x) \rangle_{p(s|x)} \\ \beta\langle F_{neq} \rangle_{p(x)} &= \beta\langle E[s, x] \rangle_{p(s, x)} + H(s|x) \end{aligned} \quad (\text{A2})$$

So the difference in F_{neq} given a new signal is as follows. Notice that the marginals are lifted up to the same joint $p(s_t, x_t) = \sum_{x_{t+1}} p(s_t, x_t, x_{t+1})$, likewise for $p(s_t, x_{t+1})$.

$$\beta\langle \Delta F_{neq}[x_t \rightarrow x_{t+1}] \rangle = \beta\langle E[s_t, x_{t+1}] - E[s_t, x_t] \rangle_{p(s_t, x_t, x_{t+1})} - H(s_t|x_{t+1}) + H(s_t|x_t) \quad (\text{A3})$$

Then from the definition of dissipated work:

$$\langle W_{diss}[x_t \rightarrow x_{t+1}] \rangle := \langle W[s_t; x_t \rightarrow x_{t+1}] \rangle - \langle F_{neq} \rangle \quad (\text{A4})$$

So we subtract (A1) - (A3), and use the entropy decomposition of mutual information:

$$\beta\langle W_{diss} \rangle = H(s_t|x_{t+1}) - H(s_t|x_t) = I_{mem}(t) - I_{pred}(t) \quad (\text{A5})$$

Finally, consider this relation over an entire protocol. For this we need the average sum of non-equilibrium free energy changes during relaxation steps, which is negative since the system tends to equilibrium in expectation:

$$\langle \Delta F_{neq}^{relax} \rangle := \left\langle \sum_{t=1}^{\tau} \Delta F_{neq}[x_t; s_t \rightarrow s_{t+1}] \right\rangle \leq 0 \quad (\text{A6})$$

Which gives us the fundamental relation:

$$\beta\langle W_{diss} \rangle = I_{mem} - I_{pred} - \langle \Delta F_{neq}^{relax} \rangle \implies \beta\langle W_{diss} \rangle \leq I_{mem} - I_{pred} \quad (\text{A7})$$

Appendix B $\Sigma_{t|t}(K)$ formulation

We want to express $\Sigma_{t|t}$ as a function of K .

The error covariance is given as follows:

$$\Sigma_{t|t} := \mathbb{E}[ee^T] \quad (\text{B8})$$

First, we express the posterior error $e := x_t - \hat{x}_{t|t}$ as a function of K by relating it to the prior error $e_- := x_t - \hat{x}_{t|t-1}$. We substitute in the state estimate update, and insert the measurement model:

$$\begin{aligned} e &= x_t - \hat{x}_{t|t-1} - K(y_t - C\hat{x}_{t|t-1}) \\ e &= x_t - \hat{x}_{t|t-1} - K(Cx_t + v_t - C\hat{x}_{t|t-1}) \\ e &= (I - KC)(x_t - \hat{x}_{t|t-1}) - Kv_t \\ e &= (I - KC)e_- - Kv_t \end{aligned} \quad (\text{B9})$$

We plug this expression for e into the definition of the error covariance.

$$\begin{aligned} \Sigma_{t|t} &= \mathbb{E}[(I - KC)e_- - Kv_t][(I - KC)e_- - Kv_t]^T \\ &= \mathbb{E}[(I - KC)e_- e_-^T (I - KC)^T] - \mathbb{E}[(I - KC)e_- v_t^T K^T] - \mathbb{E}[Kv_t e_-^T (I - KC)^T] + \mathbb{E}[Kv_t v_t^T K^T] \end{aligned} \quad (\text{B10})$$

The prior error covariance is $\Sigma_{t|t-1} = \mathbb{E}[e_- e_-^T]$ by definition. Further, by assumption, the measurement error v_t and an estimate error are independent, so the cross terms $e_- v_t$ equal 0. Finally, $\mathbb{E}[v_t v_t^T] = R$, the measurement noise.

So we can rewrite as follows:

$$\Sigma_{t|t}(K) = (I - KC)\Sigma_{t|t-1}(I - KC)^T + KRK^T \quad (\text{B11})$$

Appendix C Trace first order condition

We want to solve for the K^* which minimizes the trace of the error covariance matrix. For brevity of notation let ,at first, $A := I - KC$ and $\Sigma := \Sigma_{t|t-1}$.

We start with the previously derived Joseph form of the error covariance:

$$\Sigma_{t|t}(K) = (I - KC)\Sigma_{t|t-1}(I - KC)^T + K R K^T = A \Sigma A + K R K^T \quad (C12)$$

Take the trace J , making use of expansions, cyclicity of the trace, and symmetries:

$$\begin{aligned} J &= \text{tr}(A \Sigma A^T) + \text{tr}(K R K^T) \\ &= \text{tr}(\Sigma) - \text{tr}(K C \Sigma) - \text{tr}(\Sigma C^T K^T) + \text{tr}(K C \Sigma C^T K^T) + \text{tr}(K R K^T) \\ &= \text{tr}(\Sigma) - 2 \text{tr}(K C \Sigma) + \text{tr}(K (C \Sigma C^T + R) K^T) \end{aligned} \quad (C13)$$

Redefine for brevity $A := C \Sigma C^T + R$, which is symmetric, and invertible by definiteness assumptions. Now differentiate in respect to K :

$$\begin{aligned} \frac{dJ}{dK} &= \frac{d\Sigma}{dK} - \frac{d \ 2KC\Sigma}{dK} + \frac{d \ KAK^T}{dK} \\ &= -2C\Sigma + 2KA \end{aligned} \quad (C14)$$

Set to zero and solve for K .

$$\begin{aligned} -2C\Sigma + 2KA &\stackrel{!}{=} 0 \\ KA &= C\Sigma \\ K &= C\Sigma A^{-1} \end{aligned} \quad (C15)$$

Which gives the final form of the optimal Kalman gain:

$$K^* = C \Sigma_{t|t-1} (C \Sigma_{t|t-1} C^T + R)^{-1} \quad (C16)$$

Appendix D Details on Loewner ordering

A **Loewner partial ordering** of positive semi-definite matrices A, B is such that for any vector v :

$$A \preceq B \iff B - A \succeq 0 \iff v^T A v \leq v^T B v \quad (\text{D17})$$

This induces an ordering of several scalar functionals of covariance matrices. In particular, if A and B are positive definite matrices such that $A \preceq B$, then $\text{tr}(A) \leq \text{tr}(B)$, and also $\det(A) \leq \det(B)$. See theorem 7.7.3 and Corollary 7.7.4 b) [7, pg. 471]

The general form of the posterior error covariance is:

$$\Sigma_{t|t}(K) = (I - KC)\Sigma_{t|t-1}(I - KC)^T + K R K^T \quad (\text{D18})$$

We know that the error covariance matrices Σ are positive definite, since the measurement error covariance matrix R and process noise Q are positive definite: $R, Q \succ 0$

We want to find K^* such that:

$$\forall K, \Sigma_{t|t}(K^*) \preceq \Sigma_{t|t}(K) \quad (\text{D19})$$

To decompose $\Sigma_{t|t}$, we define the following PSD matrix S , and use the formulation of K^* by trace minimization (Appendix C) :

$$\begin{aligned} S &:= C\Sigma_{t|t-1}C^T + R \succ 0 \\ K^* S &= \Sigma_{t|t-1}C^T \implies K^* = \Sigma_{t|t-1}C^T S^{-1} \end{aligned} \quad (\text{D20})$$

We can then expand $\Sigma_{t|t}$

$$\Sigma_{t|t}(K) = \Sigma_{t|t-1} - KC\Sigma_{t|t-1} - \Sigma_{t|t-1}C^T K^T + KSK^T \quad (\text{D21})$$

Completing the square using K^* from above gives:

$$\Sigma_{t|t}(K) = \Sigma_{t|t}(K^*) + (K - K^*)S(K - K^*)^T \quad (\text{D22})$$

Since $S \succ 0$, and for any matrix M , $MSM^T \succeq 0$:

$$(K - K^*)S(K - K^*)^T \succeq 0 \quad (\text{D23})$$

Therefore

$$\Sigma_{t|t}(K) - \Sigma_{t|t}(K^*) = (K - K^*)S(K - K^*)^T \succeq 0 \quad (\text{D24})$$

So we can conclude the Loewner ordering:

$$\forall K \quad \Sigma_{t|t}(K^*) \preceq \Sigma_{t|t}(K) \quad \square \quad (\text{D25})$$

Hence, the matrix $\Sigma_{t|t}(K)$ is globally minimized in the Loewner ordering at $K = K^*$, as any deviation in K from K^* strictly increases the covariance by a positive semi-definite penalty term. Because both the trace and the determinant are monotonically increasing functions with respect to the Loewner order, $\Sigma_{t|t}(K^*)$ simultaneously minimizes both of these scalar functionals. Finally, since the natural logarithm is a strictly monotonic transformation, $\arg \min_K \ln |\Sigma_{t|t}(K)| = \arg \min_K \text{tr}(\Sigma_{t|t}(K))$.

Python Code used

```
Brain Computation 86718
Final Project
Noé Kuhn
#Libraries
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import scipy as sp
import scipy.io
import numpy.linalg as LA
import warnings
from scipy.special import rel_entr

This code aims to illustrate the previous theoretical findings
T = 50          # Number of time steps
A = np.array([[0.9]]) # Environmental transition
C = np.array([[1.0]]) # Observation matrix
Q = np.array([[1.0]]) # Process noise variance (environmental noise)
R = np.array([[2.0]]) # Thermal measurement noise variance

# =====
# 2. Initialization
# =====
# True environmental state and measurements
x_true = np.zeros((T, 1, 1))
y_meas = np.zeros((T, 1, 1))

# Optimal System (Kalman Filter)
x_opt = np.zeros((T, 1, 1)) # Posterior estimate mean
P_opt = np.zeros((T, 1, 1)) # Posterior covariance (Sigma_t|t)
P_pred_opt = np.zeros((T, 1, 1)) # Predictive covariance (Sigma_t+1|t)

# Suboptimal System (Constant Gain)
x_sub = np.zeros((T, 1, 1))
P_sub = np.zeros((T, 1, 1))
P_pred_sub = np.zeros((T, 1, 1))
K_sub = np.array([[0.8]]) # A purposefully poor rule.
# 0.2 for stubborn, 0.8 for eager

# Initial conditions (Time 0)
x_true[0] = np.array([[10.0]])
x_opt[0] = np.array([[0.0]])
P_opt[0] = np.array([[5.0]])
```

```

x_sub[0] = np.array([[0.0]])
P_sub[0] = np.array([[5.0]])
# =====
# 3. Simulation Loop
# =====
for t in range(1, T):
    # --- The Environment Evolves ---
    w_t = np.random.normal(0, np.sqrt(Q[0,0]), (1, 1))
    v_t = np.random.normal(0, np.sqrt(R[0,0]), (1, 1))

    x_true[t] = A @ x_true[t-1] + w_t
    y_meas[t] = C @ x_true[t] + v_t

    # --- OPTIMAL SYSTEM using Kalman Filter) ---
    # 1. Predict
    x_pred_opt = A @ x_opt[t-1]
    P_prior_opt = A @ P_opt[t-1] @ A.T + Q

    # 2. Update (Calculate K_opt and absorb thermal measurement)
    # K_opt = Sigma_t|t-1 * C^T * (C * Sigma_t|t-1 * C^T + R)^-1
    S_opt = C @ P_prior_opt @ C.T + R
    K_opt = P_prior_opt @ C.T @ np.linalg.inv(S_opt)

    x_opt[t] = x_pred_opt + K_opt @ (y_meas[t] - C @ x_pred_opt)
    # Sigma_t|t = (I - KC) * Sigma_t|t-1
    P_opt[t] = (np.eye(1) - K_opt @ C) @ P_prior_opt

    # Record predictive covariance for thermodynamic proxy
    P_pred_opt[t] = A @ P_opt[t] @ A.T + Q

    # --- SUBOPTIMAL SYSTEM ---
    # 1. Predict
    x_pred_sub = A @ x_sub[t-1]
    P_prior_sub = A @ P_sub[t-1] @ A.T + Q

    # 2. Update (Using fixed, non-optimal K)
    x_sub[t] = x_pred_sub + K_sub @ (y_meas[t] - C @ x_pred_sub)
    # Joseph form covariance update for suboptimal gain
    I_KC = np.eye(1) - K_sub @ C
    P_sub[t] = I_KC @ P_prior_sub @ I_KC.T + K_sub @ R @ K_sub.T

    P_pred_sub[t] = A @ P_sub[t] @ A.T + Q

# =====
# 4. Thermodynamic Entropy Calculation and MSE
# =====

```

```

# Calculate MSE (ignoring step 0 to allow initial settling)
mse_opt = np.mean((x_true[1:, 0, 0] - x_opt[1:, 0, 0])**2)
mse_sub = np.mean((x_true[1:, 0, 0] - x_sub[1:, 0, 0])**2)

#  $H[x_{t+1}|s_t] = 0.5 * \ln(2\pi e) + 0.5 * \ln|\Sigma_{t+1|t}|$ 
const_term = 0.5 * np.log(2 * np.pi * np.e)
entropy_opt = const_term + 0.5 * np.log(P_pred_opt[:, 0, 0][1:])
entropy_sub = const_term + 0.5 * np.log(P_pred_sub[:, 0, 0][1:])
# =====
# 5. Plotting Results
# =====
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 8.5), sharex=True)

# Plot 1: State Tracking
ax1.plot(x_true[:, 0, 0], label="True Environmental State

(Gazelle)", color="black", linestyle="--")
ax1.scatter(range(T), y_meas[:, 0, 0], label="Noisy Visual Input",

color="gray", alpha=0.5, s=20)
ax1.plot(x_opt[:, 0, 0], label=f"Optimal Kalman Estimator", color="blue")
ax1.plot(x_sub[:, 0, 0], label=f"Suboptimal Estimator", color="red", alpha=0.7)
ax1.set_title("Implicit Model Tracking of Hidden Environment")
ax1.set_ylabel("State Position")
ax1.legend(loc='upper right')
ax1.grid(True, alpha=0.3)

# Plot 2: Thermodynamic Proxy
ax2.plot(range(1, T), entropy_opt, label="Optimal System Entropy", color="blue")
ax2.plot(range(1, T), entropy_sub, label="Suboptimal System
Entropy", color="red", alpha=0.7)
ax2.set_title("Thermodynamic Proxy: Conditional Predictive Entropy  $H[x_{t+1}|s_t]$ ")
ax2.set_xlabel("Time Step (t)")
ax2.set_ylabel("Predictive Entropy (Nats)")
ax2.legend(loc='center right')
ax2.grid(True, alpha=0.3)

# string with our MSE results
text_str = (f"Statistical Performance (Mean Squared Error):\n"
            f"Optimal Estimator: {mse_opt:.2f}\n"
            f"Suboptimal Estimator: {mse_sub:.2f}")

```

```
fig.text(0.5, 0.02, text_str, ha='center', va='bottom', fontsize=11,  
bbox=dict(facecolor='white', alpha=0.9, edgecolor='gray', boxstyle='round,pad=0.5'))  
  
plt.tight_layout(rect=[0, 0.08, 1, 1])  
plt.show()
```